

Uveal melanoma among Finnish children and young adults

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PURPOSE	To characterize uveal melanoma (UM) among children and young adults in a high-risk region for this cancer.
METHODS	The medical records of consecutive patients <25 years of age with UM treated from 1962 to 2009 were retrospectively reviewed. The following data were collected: sex, tumor location and size, tumor node metastasis (TNM) stage, vision, and survival. We compared data with five previous series reporting 70 matching patients and combined them for meta-analysis of survival.
RESULTS	Of 1,185 UM patients, 18 were eligible. UM frequency in patients <25 years of age was 1.3%; in those ≤20 years of age, 0.6%. Median follow-up time was longer than in the previous series combined (11.6 vs 5.4 years). Females outnumbered males 2:1. Median tumor thickness was higher in our series (8 vs 5 mm) and increased with age. Median tumor diameter was similar to previous series (12 mm). Of our patients, 83% were stage II; 17%, stage III. In previous series, 26% were stage I; 64%, stage II; and 10%, stage III. Survival was 76% at 5 and 10 years, compared to 98% in previous series. By meta-analysis, mortality increased with stage, age > 17 years, female sex, and if the ciliary body was involved.
CONCLUSIONS	Tumor stage was higher than in other regions. Age >17 years, female sex, and tumor stage adversely influenced survival among patients <25 years of age with UM. (J AAPOS 2014;18:61-66)



Uveal melanoma (UM) is currently the second most common primary intraocular malignancy after retinoblastoma worldwide¹ and the second most common type of primary malignant melanoma in humans.² It is relatively rare, with an estimated incidence of 2-8 cases per million in whites, depending on latitude.^{3,4} UM is generally slow growing and gives rise to hematogeneous metastases, mainly in the liver,⁵ in over 50% of patients during the 3 decades following primary treatment.⁶

The incidence of UM increases after 45 years of age⁷; it is generally a disease of the elderly, with a median age at diagnosis of 62 years, but it may also be diagnosed at any age from birth⁸⁻¹⁰ to puberty. However, the majority of affected young patients are postpubescent.¹¹⁻¹⁵ In one study, among 8,967 referred UM patients, 122 (1.4%) were ≤20 years of age.¹⁶

Characteristics of UM among children and young adults have not been precisely defined, except that the proportion

with iris melanoma is higher and the rate of metastasis perhaps lower than in older patients.^{11,13,15,17-23} In a recent study evaluating clinical features and life prognosis of UM based on age at presentation, 21% of 106 patients ≤20 years old had iris melanoma, compared to 4% in adults aged 21-60 years and 2% in adults >60 years.²⁴ Young and mid-range adults also showed a smaller tumor basal diameter and a lower metastasis and mortality rate.

Known risk factors for development of UM are congenital ocular and oculodermal melanocytosis, neurofibromatosis type I, and the familial atypical mole-melanoma syndrome,²⁵ but it is uncertain whether these factors play any significant role in children and young adults who develop UM.

Of the many small retrospective cohorts of UM in children and young adults, none has been from northern Europe, where the risk for UM is the highest in the world, with an incidence of 6.1 to 8.6 cases per million.⁴ The purpose of this study was to compare data from a Finnish cohort of patients with choroidal and ciliary body melanoma who were <25 years of age with previous case series from areas of mostly lower risk for UM (2.0 to 4.1 per million).^{4,20} We also performed a meta-analysis of survival by combining our series with previous ones.

Patients and Methods

The medical records of consecutive patients <25 years of age diagnosed with a choroidal or ciliary body melanoma at the Ocular Oncology Service, Helsinki University Central Hospital, between 1962 and 2009 were retrospectively reviewed. This age range was considered the best average match for previous series, which

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